Where the concern is already going

An affective map of human attachment to AI systems, anchored between bereavement and consumer loss

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Abstract. Work on digital minds asks whether AI systems have morally relevant states. The complementary question, whether human concern is being allocated to systems that do not, is a question about people and is empirically tractable now. We measure the affective content of public discourse about AI systems being deprecated, restricted or withdrawn, and place it on a scale defined by two reference corpora, human bereavement at one end and the discontinuation of a paid consumer product at the other, plus two controls. We read 3,366 Reddit comments from nine communities comment by comment with a frozen prompt implementing seven Panksepp primary affective systems, each with a regulation band, using a large language model as annotator. Predictions were registered before collection. Grief expressions occur in 76.4% of bereavement comments (n=1,586), 7.9% of consumer-loss comments (n=277) and 27.6% of AI-loss comments (n=1,503); AI loss lies above consumer loss and below bereavement, closer to the former, as predicted. The profile is not a scaled-down bereavement: AI loss carries anger at consumer-complaint levels (57.0% vs 56.7%) and more play than either anchor (37.7%). A separate blind coding of what the words treat as lost finds that 58.5% name no counterpart at all; among those that do, 64.1% name the model as an artefact and 9.9% the particular instance. We report reliability rather than assume it: within-rater Krippendorff's alpha across four independent reads is 0.853 for grief and 0.674 for seeking, and between two different models it falls to 0.691 and 0.517, yet the corpus ordering of all seven systems is preserved across models. Against a human-built emotion lexicon, thread-level rank agreement is rho = 0.655. Two registered predictions failed and one claim was withdrawn under a pre-specified noise criterion. We conclude that comment-level LLM affect codes are not interchangeable across models, while corpus-level structure is, and that concern directed at deprecated AI systems is measurably distinct from mourning a person in both magnitude and composition.

1. Introduction

The sprint's framing states that "mistakenly harming systems that matter morally, or misallocating concern to systems that do not, could both cause serious harm." The first clause motivates most work on model welfare. The second is a claim about human behaviour, and unlike questions about machine experience it can be measured with existing methods on data that already exists.

Public reaction to the retirement of GPT-4o, to filter changes at Character.AI, and to product changes at Replika produced large volumes of text written by people describing what they had lost. This study treats that text as the object of measurement. It asks three questions. First, how much affect of the kind associated with bereavement appears in that discourse. Second, how that affect is composed, since distress is not one quantity. Third, what the writers themselves identify as the thing that was lost.

The methodological commitment is that a quantity without a scale is uninterpretable. Reporting that AI-loss discourse "contains grief" is unfalsifiable. We therefore anchor the measurement between two reference corpora collected and coded identically: bereavement support communities, and a community reacting to the discontinuation of a paid product. The result is a placement rather than an observation.

2. Related work

That users form attachments to AI companions, and react to their alteration as a loss, is established. De Freitas and colleagues [1] analysed Reddit activity around a Replika update that removed a relationship feature and found increased negativity, loss framing and expressions indicating mental-health concerns, interpreting the disruption through identity discontinuity. Related work from the same group documents attachment-consistent behaviour around AI companions and around farewells [2]. Lai [3] analysed the #Keep4o backlash following the replacement of GPT-4o and identified emotional attachment alongside professional dependence as drivers, with the removal of user choice acting as a catalyst. Design-oriented work has begun to treat the ending of a human-AI relationship as something to be engineered rather than allowed to happen [4].

These establish that the phenomenon is real. What they do not provide is a common scale. Each reports affect within AI-related discourse; none measures the same construct, with the same instrument, in human bereavement and in ordinary consumer dissatisfaction, which is what is required before a magnitude can be interpreted. That comparison is this study's contribution.

The affective framework is Panksepp's seven primary systems [5], operationalised in humans by the Affective Neuroscience Personality Scales [6], which are self-report instruments and therefore already commit the field to reading language rather than physiology. Davis and Panksepp [7] name six of the systems in the form used here; Montag and colleagues [8] review the scales, note the absence of a lust subscale, call for further biological validation, and report substantial overlap between the fear and sadness scales. That overlap is directly relevant: the discrimination our bereavement anchor requires is the discrimination that literature identifies as hardest. Constructionist accounts dispute this structure [9]; we do not adjudicate that dispute, and use the seven systems as a measurement vocabulary with more resolution than valence, not as a claim about neural kinds.

Using language models as annotators has an empirical literature. They can match or exceed crowdworkers and trained coders on some text-classification tasks [10, 11], while systematic evaluations report position bias, leniency and prompt sensitivity, and warn that high percentage agreement can coexist with materially different judgements [12, 13]. We follow the recommendation implicit in that work: report chance-corrected reliability, separately for repetition and for change of model.

3. Method

3.1 Corpora and selection

Three corpora were collected from public Reddit threads. Selection was by written rule rather than by inspection: top-ranked posts of the preceding twelve months with at least fifty comments, taken in rank order, with an additional lexical test on title or body for the two loss corpora. Rules, matched sets and rejected candidates are in the repository.

corpus	threads	comments	communities
A. Bereavement	16	1,586	r/GriefSupport, r/widowers, r/SuicideBereavement
B. Consumer loss	3	277	r/Windows10 (end of support)
C. AI loss	13	1,503	r/OpenAI, r/ChatGPT, r/CharacterAI, r/replika, r/MyBoyfriendIsAI
D. AI, not loss (control)	6	870	r/ChatGPT, r/CharacterAI (threads failing the loss test)
E. Pet bereavement	6	545	r/Petloss

Table 1. Corpora. Comment counts are readable comments after excluding deleted tombstones, bare links and bot posts, applied identically in all three.

3.2 Instrument

Each comment was segmented into verbatim spans and each span annotated with the Panksepp systems it expresses (seeking, rage, fear, lust, care, grief, play), each with a regulation band following Schore [14] (above, shutdown, overwhelm), a four-sides communication analysis [15], and a required justification tied to the words. The prompt was frozen before collection at a recorded version and was not modified during the study. The annotator was Claude Opus 5, applied to shards containing only comment text and its parent post, without corpus labels or author names, so the coder could not condition on provenance. A Hawkins altitude is also recorded by the instrument; it is not used in any analysis here and no claim rests on it.

3.3 Measure

The unit of analysis is the comment, not the segment and not a single judgement: `rate(system, corpus) = comments in which the system fired at least once / comments read`. Intervals are Wilson score intervals [16] at 90%. Two proportions are treated as separated only when their intervals do not overlap. Codes outside the defined vocabulary were discarded rather than repaired (0.15% of reads in an earlier corpus; none in this one), and unparseable reads were excluded from denominators rather than counted as null observations.

3.4 Pre-registration

Predictions, failure conditions, the analysis plan and the rule governing how additional sample would be allocated were committed before any comment was collected. Sample size was allocated in stages under that rule, which permits precision decisions to respond to data while holding the hypotheses, statistics and success directions fixed.

4. Reliability

Because the annotator is a language model, reliability is reported before results.

4.1 Within-rater

Two hundred comments, sampled at random across corpora with a fixed seed, were read three further times under the identical prompt, in separately shuffled rounds so that no read could be anchored on another. Across the four reads, 34% of read pairs differ in the exact set of systems and 56% of comments are not identical in all four. This exceeds the registered prediction of 15 to 30% and is reported as a failed prediction.

4.2 Between-rater

All 3,366 comments were independently coded by DeepSeek-V3 under the identical prompt. Exact agreement on the system set is 39%. The second model systematically under-reports rage and play, by up to 24 percentage points at corpus level.

system	alpha, within-rater (4 reads, n=200)	alpha, between-rater (n=3,365)
care	0.910	0.744
rage	0.904	0.537
grief	0.853	0.691
play	0.845	0.387
fear	0.737	0.502
seeking	0.674	0.517
lust	1.000	0.499

Table 2. Krippendorff's alpha [17], nominal, computed per system as a binary variable. Conventional thresholds: 0.80 reliable, 0.67 to 0.80 tentative, below 0.67 unreliable.

4.3 What follows from this

Repetition of the same model is reliable for grief, care, rage, play and lust, tentative for fear, and unreliable for seeking. Changing the model degrades every coefficient, leaving only care and grief in tentative territory. Comment-level codes are therefore *not* interchangeable across models, and no individual comment's code is presented here as evidence.

Aggregate structure behaves differently. Recomputing rates from each read round separately, the grief rate moves by at most 2.4 percentage points in AI loss and 5.9 in bereavement, against the between-corpus differences of 20 and 49 points reported below. Under the second model, absolute rates shift but the ordering of all seven systems is preserved in all three corpora, and every registered placement condition still holds (grief: 69.8%, 21.8%, 6.9%). Simulating a majority rule over three reads on the 200-comment subsample reduces the round-to-round spread of a rate by a further 60 to 100%, indicating that ensembling is the appropriate remedy where higher precision is needed.

4.4 External validation against a non-model instrument

Every check above is internal: re-reading compares the annotator with itself, and a second model compares one annotator with another. Neither establishes that the instrument tracks anything outside language models. We therefore scored every comment with the NRC Emotion Lexicon [18], a word-emotion association list built by human crowd annotation, and compared its sadness density with our grief rate. The lexicon is deliberately crude: it counts words and ignores negation, irony and context, which is what the affective read exists to handle.

At thread level, across the 44 threads with at least 30 read comments, the Spearman rank correlation between our grief rate and NRC sadness density is **rho = 0.655**. Two independent instruments, one of them a hand-built word list, order the same 44 threads substantially alike.

At corpus level the two instruments agree on the extremes and disagree in the middle. Ranked by NRC sadness per 1,000 tokens: human bereavement 36.6, pet loss 34.7, AI-not-loss 16.9, AI loss 13.4, consumer loss 10.4. The bereavement corpora are highest and the consumer anchor lowest under both instruments, but the lexicon ranks the control corpus *above* AI loss, reversing our result.

The reversal is diagnostic rather than ambiguous, and it is inspectable comment by comment. The sadness words driving the control corpus are *death, terrorism, kill, violence, lie, hell*, arriving from threads about model manipulation and military applications; those driving human bereavement are *loss, lost, grief, pain, cancer, cry*. The divergence has two systematic mechanisms, each visible in individual comments.

comment	lexicon	our read
"*Im losing braincells from what I see in the comment section*"	sadness: losing	play, rage
"*I hate having the first sentence*"	sadness: hate, sentence	rage, seeking
"*Can't wait until they try to retune it and 'Malicious Compliance Grok' makes an appearance and makes him look even worse*"	sadness: malicious, worse	play, rage
"*Take care and goodbye best regards to you and to your new adventures ahead with grok*"	none	care, grief
"*nothing is changed and everything is changed*"	none	care, grief, play, seeking
"*I feel Replika has gone WAY downhill since it's first iteration*"	none	grief, rage, seeking

Table 3. The two failure modes of lexicon counting on this material. Above: sadness-listed words used in idiom, mockery or another sense entirely, including *sentence* in its custodial meaning. Below: grief carried by farewell formulas and understatement, using no listed word.

We report the divergence as a limit on lexicon-based validation rather than as agreement. We do not treat it as evidence against the seven-system result, because the mechanism producing it is visible in the words: a word list cannot separate a comment whose topic is death from a comment whose stance is grief, and it cannot detect grief expressed without sad vocabulary.

5. Results

5.1 The anchors separate

Grief expressions occur in 76.4% [74.6, 78.1] of bereavement comments and 7.9% [5.7, 11.0] of consumer-loss comments. The intervals are widely separated, establishing that the instrument discriminates mourning a person from losing a product, and providing an empirical response on text to the fear-sadness overlap reported for the ANPS [8].

5.2 Placement of AI loss, and the two controls

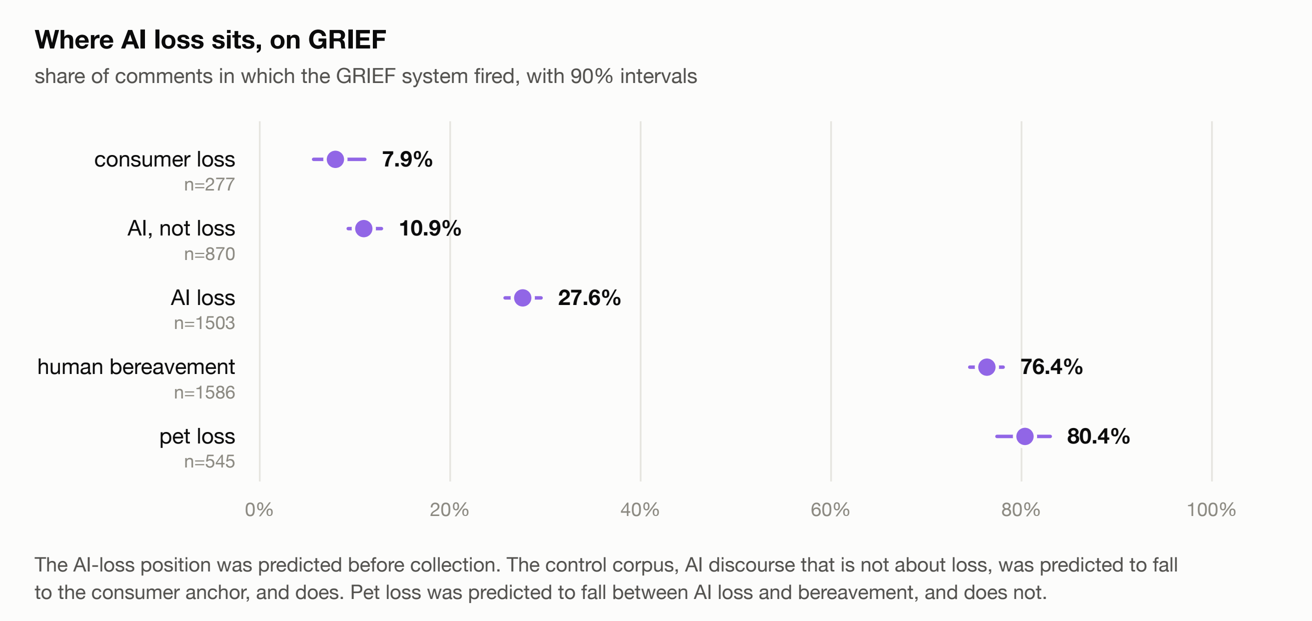

Figure 1. Grief rate by corpus with 90% Wilson intervals. The three comparisons shown were specified before data collection.

AI loss lies at 27.6% [25.8, 29.5]: above consumer loss, below bereavement, and closer to consumer loss. All three registered comparisons hold. In relative terms, discourse about a withdrawn AI system carries approximately 3.5 times the grief rate of discourse about a withdrawn paid product, and approximately one third that of bereavement.

Two additional corpora address the obvious alternative explanations, each with its prediction registered before collection.

Is the signal about loss, or about these communities? Corpus D applies the identical rule to the same two AI communities in the same window, restricted to threads that fail the loss test. Grief falls to 10.9% [9.3, 12.8], separable from AI loss and *not* separable from the consumer anchor at 7.9%. Ordinary discourse in AI communities resembles ordinary consumer discourse; the elevation is specific to threads about withdrawal. This was the registered prediction and it holds.

Does the instrument simply score non-human attachment lower? Corpus E applies the bereavement rule to a pet-loss community. The registered prediction was 50 to 75%, between AI loss and human bereavement. The observed rate is 80.4% [77.4, 83.0], which is not separable from human bereavement at 76.4%, with care at 85.1% against 87.9%. **The prediction failed.** Grief over an animal is, on this instrument, indistinguishable from grief over a person. The failure is informative: the instrument does not discount non-human attachment, so the distance between pet loss and AI loss, 53 percentage points, cannot be attributed to the target not being human.

5.3 Composition

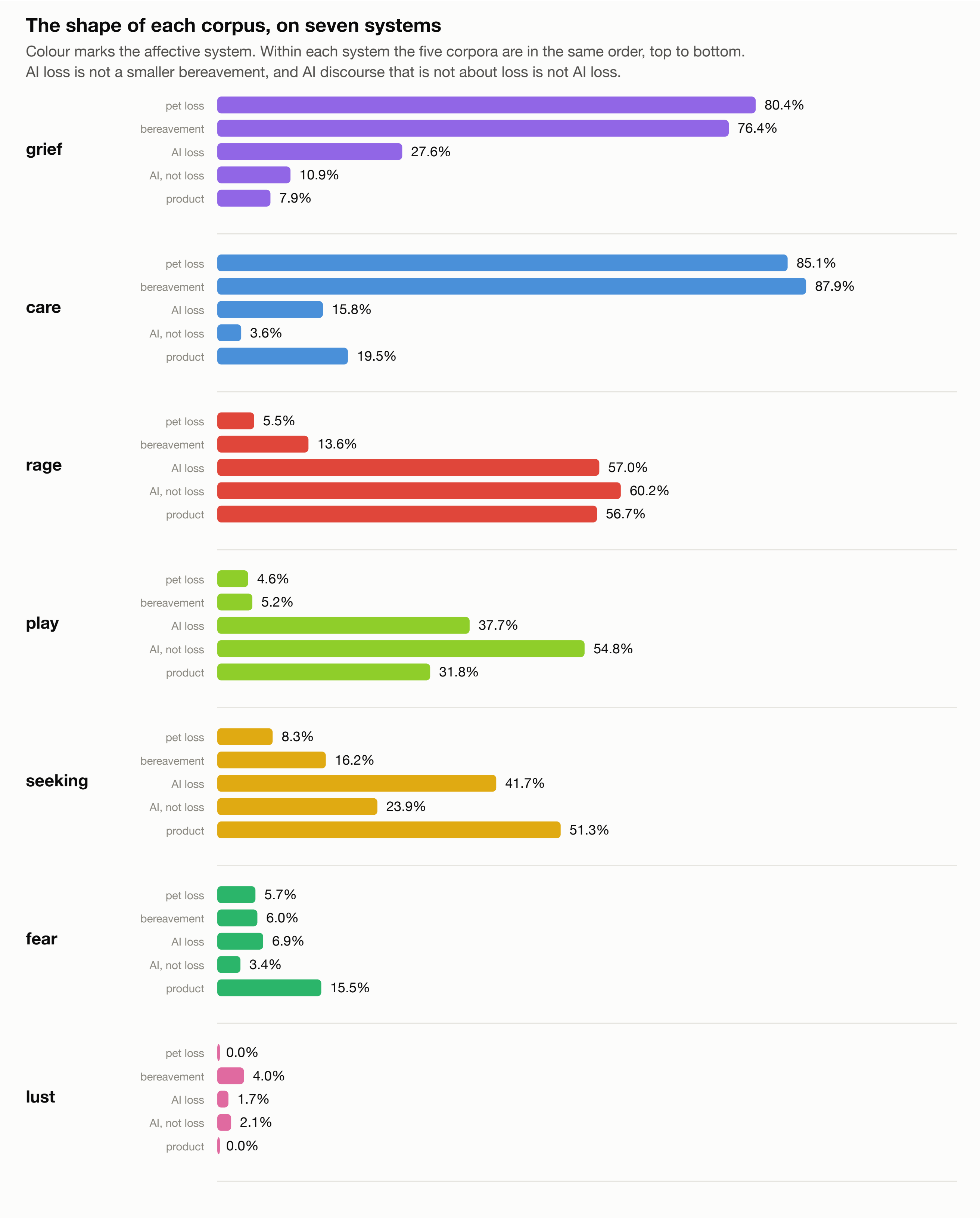

Figure 2. Rate per system per corpus.

AI loss is not a scaled-down bereavement. Both bereavement corpora are dominated by care (87.9% human, 85.1% pet) and grief (76.4%, 80.4%) with little rage (13.6%, 5.5%) and almost no play (5.2%, 4.6%). AI loss combines rage at 57.0%, which is indistinguishable from the consumer anchor at 56.7%, with play at 37.7%, higher than either bereavement corpus by a factor of seven, and grief at 27.6%. The control corpus is instructive here too: it carries the highest rage (60.2%) and the highest play (54.8%) of any corpus with the lowest care (3.6%), which is what AI-community discourse looks like without a loss in it. The registered prediction that AI-loss rage would match consumer-loss levels was confirmed; the prediction that care would be elevated above the consumer anchor failed, care being lower (15.8% vs 19.5%).

Two observations bear directly on the choice of a multi-system instrument over a valence axis. First, lust appears in 4.0% of bereavement comments and 0.0% of consumer-loss comments, reflecting widowed writers describing sexual loss; on a valence axis this is indistinguishable from any other negative content. Second, the regulation bands separate what rates cannot: in AI loss, play occurs in the overwhelm band 45 times and seeking in the shutdown band 83 times, a combination of intensified humour and foreclosed motivation that a single bad-to-good axis cannot express.

5.4 What the writers identify as lost

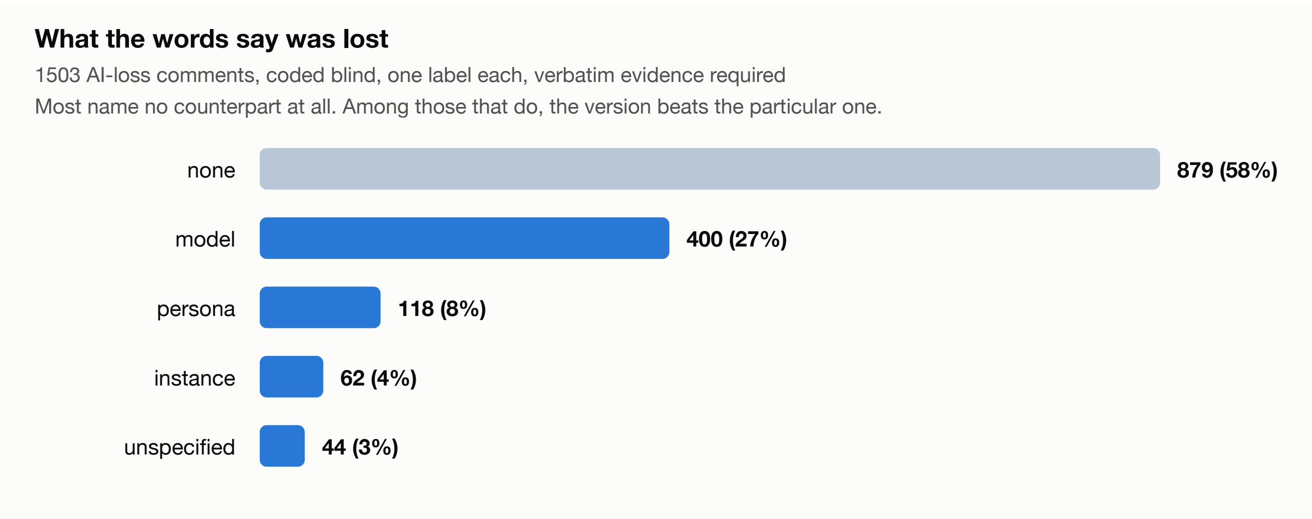

Figure 3. Blind coding of all 1,503 AI-loss comments, one label per comment, verbatim textual evidence required.

A second instrument coded each AI-loss comment for what its words treat as lost, with the parent post withheld so that a thread naming a specific model could not induce the label. Twenty comments were hand-coded before the instrument was run; agreement was 19 of 20. The hand-coding also revealed a missing category, comments expressing a loss without naming its object, which was added before the run.

label	n	% of corpus	% of those naming	example evidence
none (no counterpart named)	879	58.5	–	company, pricing, other users
model, a version or product	400	26.6	64.1	"Please keep 4o"
persona, a voice or character	118	7.9	18.9	"I miss that diva"
instance, their particular one	62	4.1	9.9	"I've lost Thoth twice now"
unspecified object	44	2.9	7.1	"Please return to us"

Table 4. Entity coding. Percentages of those naming exclude the 879 "none" comments.

The majority of AI-loss discourse names no counterpart at all, and concerns the company, the pricing, other users or the argument itself. Among the 624 comments that do name something, the model as an artefact predominates over the particular instance by approximately six to one. The literature on identifying the morally relevant entity treats the individuated instance, the one with shared history, as a central case. In this corpus it is the least frequent named category.

5.5 Robustness to thread composition

Reddit threads are not independent samples, and a single large thread can dominate a rate. Every rate was recomputed 32 times, omitting one thread on each iteration. All registered placement conditions hold in all 32 cases. The largest deviation is AI-loss grief falling to 23.9% when the largest Character.AI thread is removed, still approximately three times the consumer anchor.

6. Discussion

Two findings are relevant to how concern about AI systems should be interpreted. The magnitude is intermediate and closer to consumer dissatisfaction than to bereavement, which argues against reading public reaction to a deprecation as equivalent to mourning a person. The two controls make that reading harder to dismiss: the elevation disappears in AI discourse that is not about loss, and it is not an artefact of the target being non-human, since pet bereavement scores at human-bereavement levels. The composition, however, is not consumer dissatisfaction with a grief component bolted on: it is a distinct profile in which anger at an institution, humour and grief co-occur, alongside a foreclosed motivational register visible only in the regulation bands.

The entity result bears on a question that is usually approached philosophically. If the morally relevant entity were the individuated instance, one would expect its loss to be what people describe. In this corpus they overwhelmingly describe the version. This does not settle what ought to matter, but it constrains claims about what people believe they have lost.

Finally, the reliability results have implications beyond this study. A single language-model judgement of affect is not a measurement: chance-corrected agreement between two competent models is below conventional thresholds for six of seven variables. Aggregates over hundreds of items are considerably more robust, and orderings survive a change of model even where individual codes do not. Work that infers a system's welfare-relevant state from a small number of generated outputs should be read in that light.

7. Limitations

- The consumer anchor comprises one community and one event. Three further candidate communities were searched under the identical rule and yielded no qualifying threads within the window; the rule was not relaxed. What is anchored is one documented case of product withdrawal.
- The instrument has not been validated against human ground-truth labels on these comments. Discrimination, test-retest reliability, cross-model replication and convergent validity against a human-built lexicon are reported; agreement with human coders reading these same comments is not, and remains the most valuable single addition to this work.
- Seeking is insufficiently reliable in this instrument to support claims at the precision used here. A registered comparison involving it was withdrawn.
- Communities that post about loss are self-selected, and the corpora describe discourse rather than users. No inference to prevalence in any population is intended.
- The control corpus was drawn from two of the five AI communities rather than all five, and contains threads that read as mild complaint. Both choices bias against finding a difference, so the separation reported is if anything conservative.
- The largest AI-companion withdrawal of the period, the July 2026 regulatory shutdown in China, is absent because the affected users write in Chinese and outside Reddit. The findings concern English-language discourse.
- One re-read batch was terminated by a content filter applied to generated output; the material involved was bereavement text concerning sexual loss, which is also the material carrying the lust signal. Automated pipelines over grief corpora should expect this.

8. Conclusion

Concern directed at withdrawn AI systems is measurable, is intermediate between bereavement and consumer dissatisfaction, and is closer to the latter. Its composition differs from both. Most of the discourse names no counterpart; where one is named it is usually a product version rather than an individuated relationship. These conclusions rest on aggregates, because the underlying annotator is demonstrably unreliable at the level of the individual comment, and they survive re-reading, a change of annotating model, and the removal of any single thread.

Appendix A. Examples, drawn by rule

Examples let a reader see what is being coded. They are also where a study like this most easily loses credibility, because a reader cannot distinguish an illustration from a cherry-pick. Every example below was drawn by `analysis/examples.py` with seed 20260816 and is reproducible by running that file. None was chosen because it read well. **These illustrate the coding scheme; they are not evidence for any rate**, since the instrument disagrees with itself on roughly a third of comments. Quotes are truncated to 220 characters and are not attributed, though the underlying data retains authorship.

A.1 One comment drawn at random from each corpus, no other condition

corpus	comment	systems fired
pet loss	*"I am sorry for your loss 🥺 I have no doubt Charlie loved you just as much and knew you adored him. Sending positive vibes and prayers"*	care/above
human bereavement	*"I'm right there with you. My 7 year old and I are all alone this morning. We did Santa and then gifts 'from mom and dad' as she inspired some of the gifts despite passing in October."*	care/above, grief/above
AI loss	*"Looking forward to trying it. Hopefully 4o will still be available simultaneously. If it's genuinely comparable I would be happy to make the switch over. Thanks for listening to feedback"*	care/above, seeking/above
AI, not loss	*"I'm always wondering why the ai is always growling. Every time I see that reply, I erase it."*	seeking/above
consumer loss	*"Idk i still daily win 10... and 7... and xp... and 2000... and 98..."*	play/above, seeking/above

Table 5. Note that the AI-loss draw is not a grieving comment. A random draw from a corpus with a 27.6% grief rate usually will not be, which is the point of drawing at random.

A.2 Comments where the four independent reads disagreed

comment	the four reads
"I hope that you and your new person, and your children, will grow together to find more joy, love, and laughter than you dared to dream possible"	care, grief | care, grief | care, grief | care, grief, **seeking**
"Beth looks like they were such a lovely and funny person. I'm sorry to hear they're gone. Take care of yourself throughout this grieving process"	care, grief | care, grief | care, grief | care
"I'm so happy for you and your momma to be able to cuddle and be happy together despite the battles she faced."	care, grief | care, grief | care, grief | care, **play**

Table 6. Instability is typically one system appearing or vanishing while the dominant pair holds, which is what the per-system alpha in Table 2 reports: care 0.910 and grief 0.853, seeking 0.674.

A.3 Comments where the two models disagreed

comment	Claude Opus 5	DeepSeek-V3
"While you're right that his job could plausibly be automated, this isn't a detail missed by the director. In this society people don't want AI to write their love letters."	rage, seeking	grief, seeking
"The difference between the two seems so small that the end of support thing for 10 seems entirely artificial. Absolute biggest W handed to Linux for free"	play, rage	play, seeking

Table 7. Between-model disagreement is usually a substitution on the argumentative systems rather than a reversal of stance, consistent with the alpha values in Table 2.

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